I want to create a minecraft mod where creepers destroy energy infrastructure.

That's a solid mod concept—very "Minecraft with consequences." You can turn creepers into a gameplay hazard specifically for power systems, which opens up a lot of interesting engineering-style survival gameplay.

Here's a practical way to structure it.

Core idea

Creepers gain a new behavior: they preferentially target or are attracted to **energy infrastructure blocks**, and explosions cause **specialized damage to power networks**, not just terrain.

Think:

- Normal creeper → terrain damage
- Your modded creeper → *grid disruption + cascading failures*

Step 1: Choose your modding platform

Pick one:

- **Fabric** (lighter, faster updates, more manual work)
- **Forge / NeoForge** (more built-in systems, better for complex mechanics)

If you're new, Forge/NeoForge is usually easier for systems like block power networks.

Step 2: Define "energy infrastructure"

You need a clear set of blocks that count as infrastructure. For example:

- Power generators (solar panels, coal generators, reactors)
- Power storage (batteries, capacitors)
- Transmission (cables, relays, transformers)

If you don't want to build a full tech mod, you can:

- Tag existing mod blocks via a config system

- Or define a small vanilla-friendly system (redstone-based “energy blocks”)

Step 3: Creeper behavior changes

You have a few interesting options:

Option A — Attraction system

Creepers are “drawn” to energy blocks:

- Scan nearby chunks for tagged blocks
- Pathfind toward them if within range
- Prefer staying near energized structures

Option B — Targeted detonation

On fuse start:

- Creeper checks if it’s near energy blocks
- If yes, it “locks in” explosion direction/damage bonus toward them

Option C — Specialized “Energy Creeper”

A variant mob:

- Spawns near power networks
- Slightly more intelligent pathfinding
- Maybe glows when near high-energy areas

Step 4: Explosion effects (this is the fun part)

Instead of just breaking blocks:

1. Power network disruption

- Disable adjacent energy blocks for X seconds
- “Short-circuit” effect (blocks stop working temporarily)

2. Chain reaction

- Exploding generator can:
  - Overload nearby cables
  - Cause secondary explosions or shutdown waves

3. Data/power loss mechanic (optional)

- Stored energy is partially lost when hit
- Or gets “leaked” into the environment

Step 5: Detection system (important for realism)

You'll need a way to identify energy infrastructure:

- **Block tags** (recommended):
  - `#energy_generator`
  - `#energy_storage`
  - `#energy_conduit`

Or:

- Interface-based system (e.g., `IEnergyBlock`)

Step 6: Balance mechanics

Otherwise creepers become unfair:

- Only affect powered or active blocks
- Limited detection range (e.g., 16–32 blocks)
- Higher spawn rate near industrial bases (optional difficulty scaling)
- Players can use "shielding blocks" or insulation blocks

Step 7: Implementation outline (high level)

In code terms (Forge-style conceptual):

- Hook into Creeper AI goal system:
  - Add `MoveToEnergyGoal`
- On explosion event:
  - Scan AABB for energy-tagged blocks

- Apply custom effects:
  - block state changes
  - tile entity energy loss
  - redstone interruption / mod energy API hooks

Step 8: Expansion ideas

If you want to make it really interesting:

- "EMP creeper" variant that disables redstone temporarily
- Lightning-charged creepers that spawn near power grids
- Player-built "defense systems" (turrets, shields)
- Environmental storytelling: abandoned industrial ruins from creeper attacks

Key design question you should decide next

Do you want this to be:

1. A **hard survival challenge mod** (creepers are threats to infrastructure)
2. A **tech mod integration** (works with energy systems like FE/RF from other mods)
3. A **standalone energy system mod** you build from scratch

If you tell me which direction you want, I can help you design:

- the exact class structure
- event hooks for Forge/Fabric
- or even a starter mod template layout

It's an integration to existing tech mods. I want to spawn creepers more frequently, and in groups and for them to explode and destroy infrastructure.

Good—this is much more doable if you treat it as an **"ecosystem modifier" mod** rather than reinventing tech systems. You're basically building a hostile pressure layer on top of existing energy mods (Forge Energy / FE ecosystem mods).

Here's a clean way to design it so it actually works with other tech mods instead of fighting them.

1. High-level design (what your mod actually does)

Your mod has three responsibilities:

1) Modify creeper spawning

- More frequent creepers
- More group spawns ("packs")
- Higher spawn rates near player-built tech infrastructure

2) Detect "energy infrastructure"

- Works with other mods via common APIs (Forge Energy / capability system)
- Also optionally via block tags for modded blocks that don't expose energy cleanly

3) Modify creeper explosion behavior

- Prioritize damaging energy storage + generators + cables
- Cause "cascade failures" in nearby energy networks (optional but powerful)

2. Make it compatible with tech mods (critical step)

Most tech mods use:

- Forge Energy (FE) capability
- Or wrapper systems like RF-style APIs

You should detect energy blocks using:

Option A (best): Capability check

- Check if a block entity has energy capability
- If yes → it is "infrastructure"

Option B (fallback): Tags

Let users configure:

JSON

```json
"energy_blocks": [
  "thermal:energy_cell",
  "mekanism:induction_matrix",
  "#forge:energy_cables"
]
```

3. Creeper spawn system (packs + frequency)

You'll modify spawn behavior via:

Forge event:

- `MobSpawnEvent`
- or `BiomeLoadingEvent` (for weight adjustments)

Increase spawn rate

You can:

- Increase creeper weight in monster category
- Or inject extra spawn attempts

Example logic:

- +50–200% creeper spawn weight in "industrial density zones"

Group spawning ("creeper packs")

On spawn event:

If a creeper spawns:

- Roll chance (e.g. 30–60%)
- Spawn 2–5 additional creepers nearby

Pseudo-logic:

```
onCreeperSpawn:
    if random < packChance:
        spawn N creepers within radius 3–8 blocks
```

Optional enhancement:

- Make them spawn as a "linked group" so they don't immediately scatter

Infrastructure proximity spawning (important feature)

Scan nearby chunks:

If player has:

- generators
- energy storage

- cables

Then:

- increase creeper spawn probability in that area

This creates “industrial pressure zones.”

4. AI behavior changes (optional but powerful)

You don’t need full custom AI—just add goals:

Add a “seek energy” bias

When creeper is deciding movement:

- Scan for nearest energy block entity within X radius (e.g. 24–48 blocks)
- Slightly bias pathfinding toward it

This makes creepers feel “aware” of infrastructure without being magical.

5. Explosion system (this is your signature feature)

Hook into:

- `ExplosionEvent.Detonate`

Step A: Identify affected blocks

On explosion:

- Get all blocks in explosion radius
- Filter:
  - energy storage
  - generators
  - cables

Step B: Apply specialized damage

Instead of just breaking blocks:

For energy storage:

- Drain stored energy to zero or partial loss
- Or “overload” and destroy it

For cables:

- Break connection networks (important for tech mods)
- Optionally propagate failure outward

For generators:

- Disable temporarily ("damaged state")
- Or explode into secondary blast (config option)

Step C: Cascading failure system (advanced but very cool)

If a generator is destroyed:

- Nearby energy blocks lose stability
- Chance of secondary failures increases

This creates:

> "One creeper = infrastructure blackout"

6. Config system (you NEED this for balance)

Expose everything:

TOML

```toml
[spawning]
creeper_spawn_multiplier = 2.0
pack_spawn_chance = 0.5
pack_size_min = 2
pack_size_max = 5

[infrastructure]
detect_range = 32
affects_energy_storage = true
affects_cables = true
affects_generators = true

[explosions]
energy_drain_percent = 0.75
enable_chain_reactions = true
```

7. Important balance warning

If you don't tune this carefully, you'll accidentally create:

- unplayable tech progression
- constant base destruction loops
- "never build machines" meta

So include at least one counter-system:

Examples:

- "Faraday cage" block that reduces targeting
- Reinforced cables
- Defensive turrets (optional mod synergy later)

8. Suggested implementation structure

If using Forge/NeoForge:

```
com.yourmod.creepertech/

  events/
    SpawnEvents.java
    ExplosionEvents.java

  ai/
    CreeperEnergyGoal.java

  infrastructure/
    EnergyScanner.java
    EnergyBlockRegistry.java

  config/
    ModConfig.java

  util/
    EnergyCapabilityHelper.java
```

9. Small but important design decision

You should decide this early:

Do creepers:

A) Only *destroy energy blocks when they explode near them* (safe, balanced)

or

B) Actively *hunt infrastructure* (more aggressive, much harder gameplay)

Most mods succeed better with **A + slight bias toward B**, not full aggression.

If you want, I can next help you with:

- actual Forge 1.20+ event code for spawn modification
- detecting Forge Energy capability properly
- or designing a "creeper pack AI" that doesn't break vanilla behavior

Just tell me what version you're targeting.

I have a turret mod installed as well. I have not written mods before, so I want to make something very basic.

Good choice keeping it basic—this kind of mod is *way* easier if you start with a small "rule-changing" mod instead of full AI systems.

Since you're new and already using a turret mod, the best version of your idea is:

> "Make creepers spawn more often and occasionally spawn in small groups, with a higher chance near player tech builds."

No custom mobs. No complex AI. Just event hooks + config.

What you should build first (MVP version)

Version 1 (simple, actually doable for a beginner)

You will implement only 2 things:

1) Increase creeper spawns

- Slightly raise creeper spawn weight globally (or in overworld)

2) Creeper "pack spawning"

- When a creeper spawns naturally, sometimes spawn 1–3 extra creepers nearby

That's it. No infrastructure detection yet.

This already creates:

- more pressure on bases
- synergy with turret mods (they get more targets)
- “creeper waves” feeling

Step-by-step plan (Forge / NeoForge style)

I’ll assume you’re using a modern version like 1.20+ Forge/NeoForge.

Step 1: Setup mod environment

Install:

- Java JDK 17 or 21 (depends on MC version)
- IntelliJ IDEA (recommended)
- Forge or NeoForge MDK (mod development kit)

You basically:

1. Download MDK
2. Import into IDE
3. Run “runClient” once

Step 2: Make a basic mod file

You’ll have a main class like:

- `CreeperChaosMod.java`

Nothing fancy yet.

Step 3: Hook into mob spawns

You want this event:

- `MobSpawnEvent.FinalizeSpawn` (Forge)
  or equivalent spawn event in NeoForge

This lets you react when mobs appear.

Step 4: Add creeper pack spawning (core feature)

Concept:

> When a creeper spawns → sometimes spawn more creepers nearby

Logic:

- If entity is Creeper
- Roll random chance (like 30%)
- Spawn 1–3 extra creepers within 2–5 blocks

Example pseudocode (very close to real logic)

Java

```java
if (entity instanceof Creeper creeper) {

    if (random.nextFloat() < 0.35f) {

        int extra = 1 + random.nextInt(3);

        for (int i = 0; i < extra; i++) {

            Creeper newCreeper = EntityType.CREEPER.create(level);

            double x = creeper.getX() + (random.nextDouble() - 0.5) * 4;
            double y = creeper.getY();
            double z = creeper.getZ() + (random.nextDouble() - 0.5) * 4;

            newCreeper.moveTo(x, y, z, random.nextFloat() * 360F, 0);

            level.addFreshEntity(newCreeper);
        }
    }
}
```

That alone gives you “creeper squads.”

Step 5: Make spawn rate slightly higher (optional but easy)

You can either:

Option A (simple, no biome editing)

- Occasionally spawn an extra creeper when any hostile mob spawns

Option B (cleaner)

- Modify spawn weights in biome loading event

But for beginners: **don’t touch biome spawn tables yet**.

Step 6: Make it feel good with turrets

Since you already have a turret mod installed:

Your mod automatically creates:

- more combat situations
- more defense gameplay
- more base pressure

You don’t need to integrate directly.

But later you *could*:

- increase pack size near turret-heavy areas
- or reduce spawn near “safe zones”

What NOT to do yet (important)

Avoid these until you’re comfortable:

- custom creeper AI goals ❌
- energy system detection ❌
- explosion modification ❌
- cross-mod APIs ❌
- chunk scanning systems ❌

These are where most beginners get stuck.

A good “Phase 2” after this works

Once pack spawning works, then you upgrade:

Phase 2 ideas:

- Creeper spawn scaling near player bases
- More packs at night
- “siege waves” every X nights
- interaction with turrets (optional targeting logic)

Minimal design summary

Your first working mod is:

> "Creepers sometimes spawn in groups instead of alone."

That's already fun, noticeable, and compatible with everything.

If you want next step help, I can walk you through:

- setting up Forge/NeoForge project step-by-step
- where exactly to paste this code in a real mod project
- or how to test spawn events quickly in-game (so you don't waste time guessing)

Just tell me what version of Minecraft you're targeting.

It is necessary that the creepers are actively destroying infrastructure.

That requirement pushes this from "spawn tweak mod" into a **targeted behavior mod**, but you can still keep it beginner-friendly if you *avoid full AI rewrites* and instead do a simple rule:

> "If a creeper is near energy infrastructure, it tries to go there before exploding, and its explosion prioritizes those blocks."

You do **not** need custom mobs or complex systems—just two hooks:

1. detect energy blocks (from other mods via Forge Energy capability)
2. slightly steer creepers toward them + bias explosion damage

Minimal workable design (beginner-safe, but meets your goal)

You will implement ONLY 2 mechanics:

1) Infrastructure detection (lightweight)

Detect blocks that have energy storage capability (Forge Energy).

2) Creeper "bias behavior"

When a creeper is about to explode:

- check if energy blocks are nearby
- if yes → move creeper toward them (small nudge, not full AI rewrite)
- when explosion happens → target those blocks first

Step 1: Define "infrastructure"

In modded Minecraft, most tech mods expose:

- Forge Energy (FE capability)

So your rule becomes:

> A block is "infrastructure" if it has energy capability OR is tagged as energy-related.

You do NOT need to support every mod manually.

Simple helper idea

You will scan nearby blocks and check:

- does block entity have energy capability?

That's enough for:

- generators
- batteries
- machines (sometimes)
- cables (depends on mod)

Step 2: Make creepers "seek infrastructure" (simple trick)

You don't actually rewrite AI.

Instead:

Hook into tick event:

- every few ticks, for each creeper:

    - scan nearby area (say 16 blocks)
    - if energy block found:
        - slightly move creeper toward it

This is NOT full pathfinding.
It's a gentle "magnet effect."

Effect:

- creepers drift toward power systems
- turrets still function normally
- behavior feels intentional but not over-engineered

Step 3: Explosion targeting (this is the key feature)

Hook:

- `ExplosionEvent.Detonate`

When creeper explodes:

Do this:

1. Get all blocks in explosion radius
2. Filter:
    - energy storage blocks
    - generator blocks
    - cable blocks
3. Prioritize breaking those first (or amplify damage)

Simple version:

Instead of "random destruction," do:

> "Remove all energy blocks in radius first, then apply normal explosion."

That alone achieves your design goal.

Step 4: Make creepers more dangerous near tech (optional but important)

Add a rule:

- If creeper is near infrastructure:
  - increase explosion radius slightly OR
  - increase chance of charged creeper variant

This makes infrastructure feel "attractive" to danger.

Step 5: Minimal code structure

You only need 3 components:

```
mod/
  CreeperInfrastructureMod.java   (main mod file)

events/
  CreeperTickHandler.java         (movement bias)

  ExplosionHandler.java           (damage logic)

util/
  EnergyScanner.java              (detect FE blocks)
```

Key implementation idea (very important)

You are NOT making smarter creepers

You are making:

- **normal creepers + environmental bias system**

This is the secret to keeping it simple.

What your mod will feel like in-game

Without complex AI, you'll get:

- creepers appear in normal gameplay
- they "lean" toward machine bases
- explosions consistently hit:
  - generators
  - batteries
  - cables
- turrets get more value because attacks cluster around bases

So the gameplay becomes:

> “Your power grid attracts creeper attacks.”

Common mistake to avoid

Don’t try this early:

- full pathfinding goal rewriting ❌
- custom navigation systems ❌
- per-mod compatibility layers ❌

You don’t need them for this concept.

If you want the next step

I can help you turn this into actual working Forge/NeoForge code by walking you through:

- exact event class for creeper ticking
- how to scan nearby blocks safely (without lag)
- how to check Forge Energy capability properly
- a working explosion event handler

Just tell me your Minecraft version (e.g., 1.20.1, 1.21.x), because the code structure changes slightly.

It doesn't need to be that complex, instead of moving towards me and exploding, I just want creepers to move towards certain blocks and explode.

Good—that version is *much* simpler, and it’s actually a perfect first mod.

You’re basically making:

> “Creepers that preferentially walk toward energy-related blocks instead of the player.”

No custom AI trees, no explosion rewrites required.

Minimal design (what you’re actually building)

You only need one behavior change:

When a creeper is idle or choosing a target:

- If there is a “valid infrastructure block” nearby → set that as its navigation target
- Otherwise → behave normally (go after player as usual)

That’s it.

Step 1: Define “target blocks”

Since you’re integrating with tech mods, use Forge Energy capability:

A block counts if:

- It has an energy capability (`FE`)
- OR optionally is in a tag like `#forge:energy_storage` , `#forge:machines`

So your mod logic is:

> “Find nearest block entity that stores or handles energy.”

Step 2: Use a tick event (simple AI override)

You hook into:

- `LivingTickEvent` (Forge/NeoForge equivalent)

Every few ticks:

For each creeper:

1. If it has no current target OR random chance triggers
2. Search nearby blocks (e.g., 16–24 block radius)
3. If energy block found:
   - set that as navigation target

Step 3: Force creeper to walk toward it

Creepers already use Minecraft’s navigation system.

So you just do:

- `creeper.getNavigation().moveTo(x, y, z, speed)`

You are NOT rewriting AI goals—just occasionally giving it a new destination.

Step 4: Explosion stays vanilla

Important: you do NOT need to touch explosion code.

Creepers will naturally:

- walk to target block
- explode when close or triggered

So your mod effect becomes:

> "Creepers suicide into machines instead of chasing the player."

Simple logic flow

Every creeper tick:

```
if (no player recently hit creeper OR random chance):
    target = findNearestEnergyBlock(16-24 blocks)

    if target exists:
        navigate toward target
```

What "findNearestEnergyBlock" does

Very simple scan:

- get blocks in radius cube (16–24 blocks)
- for each block entity:
  - check energy capability
- return closest match

Stop at first good target (don't over-optimize)

Why this works well

You get:

- Creepers naturally "infecting" tech bases

- No need for custom mobs
- Works with any mod using Forge Energy
- Turrets automatically become important
- Player defense becomes meaningful

And importantly:

> You are not breaking Minecraft AI—you're just occasionally redirecting it.

Minimal class structure

You only need:

```
CreeperTechMod.java

CreeperTargetingHandler.java
EnergyBlockFinder.java
```

Important balance suggestion (very small but critical)

Without this, creepers may feel unfair:

Add at least one limiter:

- only 1 target change every 5–10 seconds per creeper

Otherwise they "jitter" between blocks.

What you are NOT doing (good news)

You avoid:

- custom mobs ❌
- explosion modification ❌
- complex pathfinding goals ❌
- cross-mod registries ❌

Result in gameplay

You'll see:

- creepers wandering into power rooms

- accidental base explosions
- turrets being triggered more often
- energy setups becoming “attractors for danger”

It feels emergent, not scripted.

If you want, next step I can show you the *exact Forge 1.20.1 code* for:

- ticking creepers safely (without lag)
- scanning nearby block entities efficiently
- and forcing navigation without breaking vanilla AI

Just tell me your Minecraft + loader version.